Scaled Graphs

One picture or one square can stand for more than 1.

Each picture stands for 2

Fish we saw

Mon	3 fish
Tue	5 fish
Wed	2 fish

Each = 2

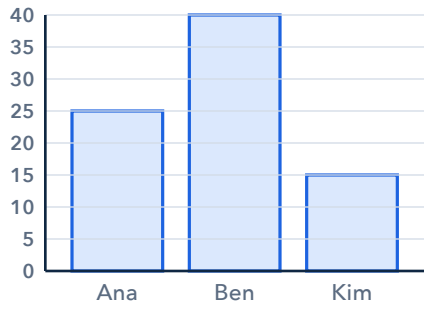
Monday has 3 fish pictures.

$3 \times 2 = 6$ fish

Count by 2s: 2, 4, 6.

A bar graph that counts by 5s

Books read



Each line is 5 more.
Ben read 40. Kim read 15.

$15 + 25 = 40$

Ben read 25 more than Kim.

Your turn!

Stars earned

Leo	4 stars

Each = 5

Leo has 20 stars

For teachers

Goal: read and draw picture and bar graphs where one picture or square stands for 2, 5 or 10 (3.MD.3), and answer how many more or less. Ask "What does one picture stand for?" before counting.

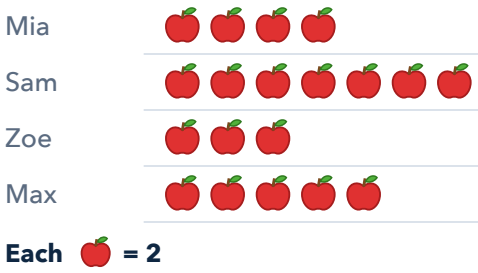
Scaled Graphs: Practice

Find what one picture or one square stands for. Then count by it.

Part 1 Each picture stands for 2 Count by 2s.

1)

Apples picked



How many apples by Mia? 8

How many more Sam than Zoe? 8

Zoe and Mia together? 14

How many in all? 38

2)

Birds at the feeder



How many birds on Mon? 12

How many more Wed than Tue? 12

Wed and Thu together? 26

How many in all? 42

Scaled Graphs: Practice (continued)

Part 2 Each picture stands for 5 or 10 Count by 5s or 10s.

3)

Stickers

Ava ★★★★★

Leo ★★★★★★★★

Kim ★★★

Jay ★★★★★★

Each ★ = 5

How many stickers for Ava? **25**

How many more Leo than Kim? **25**

Jay and Ava together? **55**

How many in all? **110**

4)

Cupcakes sold

Fri 🧁🧁🧁🧁🧁🧁

Sat 🧁🧁🧁🧁🧁🧁🧁🧁🧁

Sun 🧁🧁🧁🧁

Each 🧁 = 10

How many cupcakes on Fri? **60**

How many more Sat than Sun? **50**

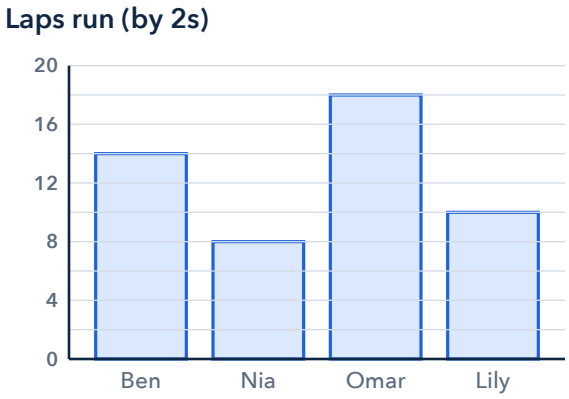
Sun and Sat together? **130**

How many in all? **190**

Scaled Graphs: Practice (continued)

Part 3 Bar graphs Each line is 2 or 5 more.

5)



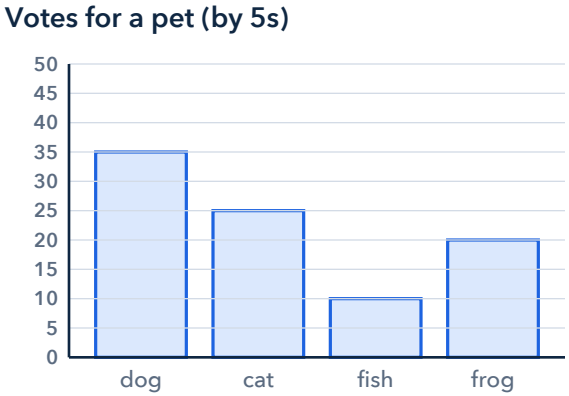
How many laps by Ben? 14

How many more Omar than Nia? 10

Omar and Lily together? 28

How many in all? 50

6)



How many votes for frog? 20

How many more dog than fish? 25

Dog and fish together? 45

How many in all? 90

Scaled Graphs: Practice (continued)

Part 4 Make the graph Draw each bar to the right line.

7) **Pages read**

Ana	30
Ravi	45
Ella	20

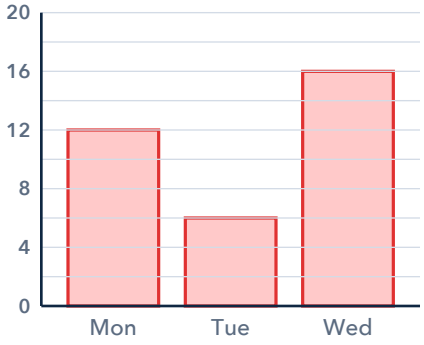
Make the bar graph (count by 5s)



8) **Shells found**

Mon	12
Tue	6
Wed	16

Make the bar graph (count by 2s)



Part 5 How many? No help this time.

9) each picture = 5: 7 pictures = 35

10) each picture = 2: 9 pictures = 18

11) each picture = 10: 6 pictures = 60

12) each picture = 5: 4 pictures = 20

13) each picture = 2: 8 pictures = 16

14) each picture = 10: 3 pictures = 30

15) each picture = 5: 9 pictures = 45

16) each picture = 2: 6 pictures = 12

Scaled Graphs: Practice (continued)

Part 5 How many? (continued)

17) each picture = 10: 8 pictures = 80

18) each picture = 5: 6 pictures = 30

Part 6 How many pictures? No help this time. How many groups?

19) each picture = 2: 7 pictures = 14

20) each picture = 5: 8 pictures = 40

21) each picture = 10: 4 pictures = 40

22) each picture = 2: 5 pictures = 10

23) each picture = 5: 3 pictures = 15

24) each picture = 10: 9 pictures = 90

Part 7 How many does it show?

25) each picture = 5: 2 pictures = 10

26) each picture = 2: 2 pictures = 4

27) each picture = 10: 2 pictures = 20

Scaled Graphs: Practice (continued)

Part 8 How many pictures?

28) each picture = 2: [3] pictures = 6

29) each picture = 10: [8] pictures = 80

30) each picture = 10: [5] pictures = 50

31) each picture = 5: [9] pictures = 45

32) each picture = 5: [6] pictures = 30

33) each picture = 5: [7] pictures = 35

34) each picture = 2: [9] pictures = 18

35) each picture = 10: [2] pictures = 20

36) each picture = 10: [6] pictures = 60

37) each picture = 5: [4] pictures = 20

38) each picture = 2: [2] pictures = 4

39) each picture = 2: [4] pictures = 8

Scaled Graphs: Practice (continued)

Part 8 How many pictures? (continued)

40) each picture = 10: 3 pictures = 30

41) each picture = 10: 7 pictures = 70

42) each picture = 5: 2 pictures = 10

43) each picture = 5: 5 pictures = 25

Part 9 Remember? Problems from earlier packets.

44) $9 \times 4 = 36$ so $9 \times 8 =$ 72

45) $500\text{ g} + 350\text{ g} =$ 850 g

46) 6 cm by 5 cm: area = 30 sq cm

47) 15 minutes after 7:50 = 8:05

48) $56 \div 7 =$ 8

49) $700 - 365 =$ 335

Scaled Graphs: Practice (continued)

Part 10 Word problems Draw a graph if it helps.

50) On a picture graph each star stands for 5. Kim has 6 stars. How many points is that?

Number sentence 6 × 5 = 30

Answer 30 points

Draw or show your work

51) Each picture stands for 2 kids. The soccer row has 8 pictures. How many kids like soccer?

Number sentence 8 × 2 = 16

Answer 16 kids

Draw or show your work

52) A bar graph counts by 10s. The red bar reaches 70 and the blue bar reaches 40. How many more is red?

Number sentence 40 + 30 = 70

Answer 30 more

Draw or show your work

53) Each apple picture stands for 10 apples. A farm picked 90 apples. How many pictures?

Number sentence 90 ÷ 10 = 9

Answer 9 pictures

Draw or show your work

54) Challenge: Each picture is 5 votes. Pizza has 7 pictures and tacos have 4. How many votes in all?

Number sentence 7 + 4 = 11, 11 × 5 = 55

Answer 55 votes

Draw or show your work

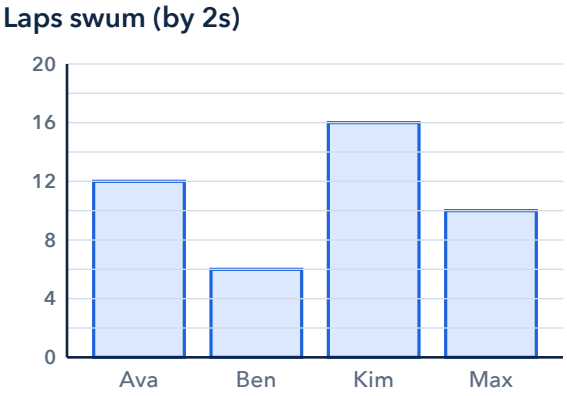
Check-In: Scaled Graphs

Show what you know. No help this time.

Mastery: 6 of 7 (80%)

Part 1 Read the graph

1)



How many laps by Max? 10

How many more Kim than Ben? 10

Ava and Max together? 22

How many in all? 44

Part 2 How many?

2) each picture = 10: 5 pictures = 50

3) each picture = 2: 5 pictures = 10

4) each picture = 2: 3 pictures = 6

5) each picture = 5: 3 pictures = 15

Part 3 How many pictures?

6) each picture = 2: 8 pictures = 16

7) each picture = 2: 6 pictures = 12